CERES

CIRCUIT-MODELLED SYNTH / USER GUIDE / 1.0.0f1

1-16 voices per layer, velocity expression and stereo Double mode.



Start playing

1. Load a sound. Create Ceres on an instrument track. Use the native patch browser to choose Init, a modern sound or a Classic Estimate.

2. Play and shape. Play softly and firmly to hear the patch's velocity response. Change the filter cutoffs, BRILLIANCE and RESONANCE, then adjust the envelopes.

3. Connect the outputs. Connect L and R to hear stereo Double patches. Mono sends the first layer equally to both outputs. Detune and Pan become editable in Double.

4. Choose voices and quality. The bank starts at four voices per layer and 1x quality. Turn the rack around to change these. Higher settings use more CPU; let held notes, pedal and release envelopes finish before a change applies.

23 sounds to begin with

Choose Init, nine modern basses, plucks and pads, or thirteen Classic Estimates. The full inventory and level guidance follow the panel reference.

Front panel

Sources and PWM	Saw, Pulse and Noise enter the filters. Sine joins the filtered signal before the amplifier. Pulse Width sets the pulse shape; PWM Rate and Depth add independent cyclic width movement.
HP and LP filters	HP removes low frequencies; LP removes high frequencies. Each has its own cutoff and resonance. Key Track moves the filter response with played pitch.
BRILLIANCE / RESONANCE	BRILLIANCE changes the overall brightness around its center detent. RESONANCE raises the shared resonance emphasis from MIN, which is the patch's neutral position. The individual HP and LP resonance controls still set the starting tone.
Filter envelope	Initial sets the starting and release target below the held level. Peak sets the attack target above it. Attack rises to Peak, Decay returns toward the held level, and Release returns toward Initial after key-up.
Volume envelope	Attack sets the rise, Decay the fall toward Sustain, and Release the key-up tail. VCF Level balances the filtered sources against the direct Sine level. Output sets the final level.
SUB modulation	The five front FUNCTION buttons select sine, down saw, up saw, square or noise. Rate sets periodic speed; pitch, filter and amplitude depths set the movement. Noise does not follow periodic Rate. Rear FUNCTION EXT selects external modulation.
Velocity	Level, Filter and Vibrato make note-on velocity change loudness, brightness and pitch modulation. Rear WAH adds velocity-scaled SUB filter modulation. The included patches use velocity; the saved Expression Source host parameter also supports channel aftertouch.
Ring modulation	Mix blends the ring sound; Speed sets its carrier rate. Depth, Attack and Decay sweep that rate. Release all keys before a new phrase to start another attack. The carrier runs continuously between notes.
Tail and Sustain I / II	Tail sets a shared minimum release time. Sustain I keeps released tails at their original pitch. Sustain II lets free tails follow subsequently assigned notes without restarting their envelopes. The MIDI sustain pedal separately holds notes.
Pitch and time readouts	Octave and Tune set pitch; the pitch wheel bends by two semitones each way. Attack is time to peak. Decay and Release use T60: nominal time for the remaining envelope difference to fall to 0.1%. Audible tail length depends on the sound.

Rear panel



ENGINE & OUTPUT	MODE chooses Mono or Double. DETUNE is the total separation between layers: 12 cents gives -6 and +6 cents. PAN sets their stereo spread; 0% centers both and 100% fully separates them. Both engines keep running in Mono, so Mono is not a CPU-saving mode.
VOICES / QUALITY	VOICES selects 1-16 per layer; release tails also occupy voices. QUALITY selects 1x, 2x or 4x internal processing. Higher quality reduces aliasing at greater CPU cost. The front status adds * while a voice/quality change is pending. Every quality reports the same 31-sample latency.
AMPLIFIER & TOUCH	LEVEL sets the common VCA level before the envelope. WAH adds touch-scaled SUB filter depth. EXPRESSION is the pedal opening when used alone. With PEDAL at CC11, the more closed of EXPRESSION and Reason's Expression controller determines the opening. OFF ignores CC11.
VOICE VARIATION	DRIFT adds free-running pitch wander; SPREAD gives the active boards a fixed tuning spread. PITCH, FILTER, ENV, ENV TIME and VCA set board-to-board variation. TEMP models an ambient temperature change since tuning. The bank includes 1 cent of drift and full board tolerances.
PORTAMENTO	OFF disables glide. PORT moves continuously to a newly assigned pitch; GLISS moves in semitone steps. RATE is measured in semitones per second, so larger values move faster.
EXTERNAL INPUT	Patch a mono signal into EXT IN, select FUNCTION EXT and raise LEVEL. The signal replaces the SUB waveform and acts through the pitch, filter and amplitude modulation amounts. EXT IN is a modulation input, not a direct audio-through path. Select any front FUNCTION button to return to an internal waveform.
AUDIO OUTPUTS	L and R carry the instrument audio. Use both for stereo Double. Live mode, detune and pan edits ramp over 20 ms. Voice and quality changes wait for all keys, pedal and release envelopes to be idle, fade the remaining passive tail, then reconfigure.

Factory bank

Init	Init
Bass	Warm Current, Pocket Funk, Night Drive
Pluck	Amber Keys, Velvet Clav, Moon Pearl
Pad	Golden Hour, Cloud Silk, Distant Lights
Classic Estimates	String I (Estimate), String II (Estimate), Brass I (Estimate), Brass II (Estimate), Flute (Estimate), Electric Piano (Estimate), Clavichord (Estimate), Harpsichord (Estimate), Guitar I (Estimate), Guitar II (Estimate), Funky I (Estimate), Funky II (Estimate), Funky III (Estimate)

Levels, CPU and saving

Patch levels. Patches store fixed output trims and retain their dynamics. Perceived loudness and peaks vary with pitch, velocity, chord density, modulation, quality and Double settings. Watch the mixer meters when changing a sound or moving from single notes to chords.

Output protection. A linked stereo sample-peak guard limits peaks near -1 dBFS with immediate attack and a 50 ms release. Strong signals and dense chords can engage it audibly. Reduce Output or the saved Output Trim if protection becomes part of the sound; sample-peak protection does not guarantee a true-peak ceiling.

Classic Estimates. These thirteen patches interpret documented preset recipes through Ceres' controls. Shared HP/LP envelope endpoints cannot reproduce separate original envelope scales or attack headroom exactly. They are estimates, not measured hardware matches. Their name suffix makes that distinction visible in the browser.

CPU and latency. Start with four voices and 1x. Increase quality when a part benefits from less aliasing and the audio engine has headroom. High polyphony at 4x can exceed the available audio budget. Lower voices or quality if needed. At 48 kHz the fixed 31-sample device latency is about 0.65 ms; the host audio buffer adds its own latency.

Save and automate. The native patch browser saves all 66 sound and rear settings. Brilliance Offset, Resonance Bus Offset, Output Trim and Expression Source are additional saved host parameters. CC11 performance messages are not stored as a patch setting. Ceres has no built-in reverb, delay or chorus.